

Electrophoretic Deposition of PbS QDs for SWIR Photodetectors

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Abstract—PbS QDs have potential for expanding of photodetectors on short-wave infrared market. An electrophoretic technique is demonstrated that improves PbS QD film morphology. The electrophoretic prepared photoconductor achieved a high responsivity than the traditional method, highlighting the potential of the electrophoretic method for next-generation near-infrared photodetectors.

Keywords—electrophoretic deposition, photoconductors, PbS QDs

I. INTRODUCTION

PbS quantum dots (PbS QDs) are a new type of semiconductor nanoparticles on the nanoscale, widely used in fields such as displays, biomedical applications, photovoltaic power generation, and lasers[1]. As a photodetector material, PbS QDs have adjustable, wide absorption spectra, and low production costs. They can cover the entire short-wave infrared (SWIR) spectrum. The broad and adjustable absorption spectrum of PbS QDs is attributed to their tunable bandgap properties. This makes photodetectors made from PbS QDs advantageous for specific wavelength detection, low cost, good environmental stability, and short response time[2]. As nanocrystals, PbS QDs need to be assembled into a thin film to exhibit macroscopic photoelectric characteristics and be made into photodetectors. The accumulation method during quantum dot film formation largely determines the spacing between quantum dots and the carrier accumulation dynamics. In other words, there is a structure-function relationship between the quantum dot film formation method and the quantum dot photodetectors [3]. Therefore, understanding the mechanism during quantum dot film formation and developing new, efficient quantum dot film formation methods will help push quantum dots into broader markets.

In this work, a simple and fast method of preparing quantum dot photoconductor using electrophoretic deposition (EPD) film

formation is proposed[4-6]. The positive and negative electrodes of the photoconductor are connected to the two ends of a DC power supply and immersed in a hexane-PbS solution for electrophoretic deposition[7]. The fabricated QD film and photodetectors are evaluated using scan electron microscopy (SEM) to analysis the surface morphology. Optoelectrical characterization is performed to analysis the responsivity of the fabricated photoconductors. The photodetector prepared in this EPD method achieved a high responsivity of 18.95 A/W at 1310 nm. It shows that the PbS QDs photoconductor prepared using EPD method shows potential as a new generation solution for near-infrared spectrum detection photodetectors.

II. EXPERIMENTAL SECTION

A. Synthesis of PbS QDs

Firstly, the PbS QDs are prepared using a low temperature, rapid injection method. Add 0.4 g of lead oxide (PbO) to a mixed solution of 8 mL oleic acid (OA) and 12 mL octadecene (ODE) as the lead precursor. Place the mixture in a flask and slowly heat it to 40°C. Under continuous evacuation and stirring, the solution eventually becomes a clear liquid. Slowly heat the mixture to 120°C and maintain the vacuum for 2 hours. The sulfide precursor is prepared by dissolving 1 mL of trimethylsilyl sulfide (TMS) into 9 mL of ODE. Set the temperature to 100°C, and when it cools to 101°C, quickly inject the sulfide precursor into the flask within 1 second, increase the stirring speed, and hold for 30 seconds. Remove the heat source and further purify the PbS solution. The prepared PbS QDs are purified by adding hexane and acetone, followed by high-speed centrifugation. The synthesized PbS QDs are characterized by transmission electron microscopy (TEM) as shown in Figure 1. The size of the QDs is measured as 5-8 nm in average.

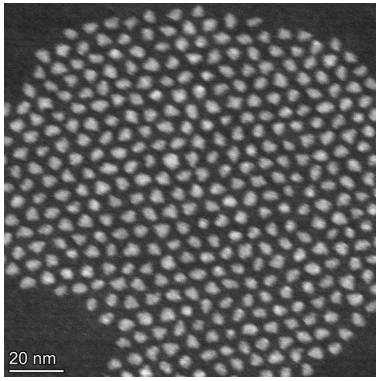


Fig. 1. TEM image of the fabricated PbS QDs.

B. EPD Process of PbS QD Film

The EPD process of PbS QD film is shown in Figure 2. Firstly, 2 ml of a 20 mg/ml hexane-PbS solution is prepared. Then, 0.6 ml of acetone is added into the hexane-PbS solution. The purpose of adding acetone is to treat the oleic acid ligands of the PbS QDs into a metastable state that is easy to be removed. Connecting the electrodes of the device to the positive and negative terminals of a DC power supply and apply a voltage of 20 V. Then, immersing the device in the hexane-PbS solution, as is shown in Figure 2(a). After two minutes, reverse the positions of the positive and negative electrodes, and perform EPD for another two minutes, as is shown in Figure 2(b). Adjust the voltage between the electrodes to 0.8 V, and transfer the device to a 10 mg/mL methanol-Tetrabutylammonium (TBAI) solution for 30 seconds for ligand exchange, as is shown in Figure 2(c). Remove the device and wash it with methanol to remove any residual TBAI solution. After the methanol has evaporated, repeat the above steps two more times.

As a comparison, another group of layer-by-layer (LBL) deposited devices is fabricated using a classic solid-state ligand exchange process discussed in previous works [8].

The working mechanism of EPD is utilizing the charged nature of PbS QDs to fabricate thin films. The electric field force will drive the PbS QDs move towards the positive and negative electrodes of the substrate and arranges the PbS QDs using the electric field force. To maximum the working area of the photoconductor, we designed and fabricated an interdigital electrode as the substrate, as is shown in Figure 3. The separation between two fingers is set as 50 μm , aiming at a higher electrical field when applying a low bias voltage.

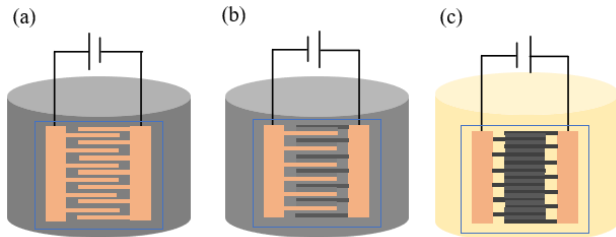


Fig. 2. Immersing the device in the hexane-PbS solution(a) and reversing the positions of the electrodes(b) and the ligand exchange(c).

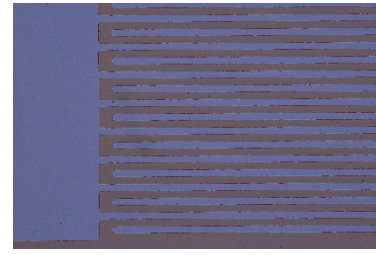


Fig. 3. The optical microscope(OM) image of the structure of the device, photoconductor, with the separation of 50 μm between two electrodes.

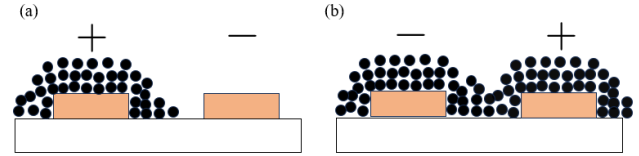


Fig. 4. The deposition of quantum dots at the positive electrode after 2 minutes(a) and the deposition of quantum dots after 2 minutes, switching the electrode(b).

During the EPD process, PbS QDs deposit on the part of the photoconductor connected to the positive electrode and grow towards the electrode gap. Therefore, after two minutes of deposition, it is necessary to switch the connections of the photoconductor to the positive and negative electrodes. This ensures that PbS QDs deposit and epitaxially grow on each electrode of the photoconductor, as shown in Figure 4.

C. Characterization of Devices

The devices are characterized using a Keithley 4200 to analyze the current-voltage curves. The devices are illuminated using a MDL-NS-1310-10mW 1310 nm nanosecond laser with a power density measured by Thorlabs PM 400 of 16 $\mu\text{W}/\text{cm}^2$. The laser is driven by a RIGOL DG1022 waveform generator.

III. RESULTS AND DISCUSSION

A. SEM Images of LBL and EPD PbS QD Film

The SEM images exhibit the surface morphology of PbS QD films. The PbS QD films using a solid-state ligand exchange (Layer-by-Layer) method is shown in figure 5(a), where stress-induced shrinkage and tearing of the PbS QDs film can be observed due to solvent evaporation. In contrast, the PbS QDs aggregated into clusters with diameters ranging from 40 to 60 nm after EPD process, forming a highly ordered arrangement without visible cracks, as is shown in figure 5(b). This indicates that the ligand exchange was more thorough, and the electric field in the EPD method maintained a shorter spacing between PbS QDs.

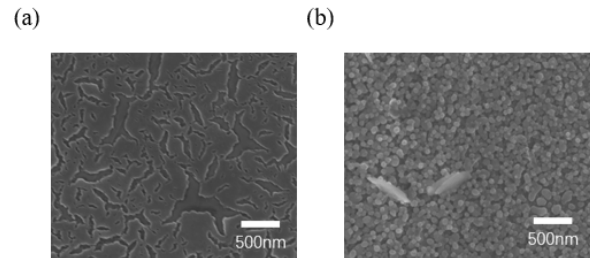


Fig. 5. The SEM image of PbS QD films fabricated using (a) LBL method and (b) EPD method.

B. I-V Characterization of Photoconductors

The I-V curve reveals the macroscopic optoelectronic characteristics of the PbS QD photoconductor. As shown in Figure 6, it can be observed that films formed using the EPD method exhibit better photoelectric separation properties. This improvement may be due to the reduced spacing between PbS QDs, allowing for a higher density of PbS QDs per unit area, which ultimately leads to the generation of more charge carriers under illumination. Additionally, the films have a lower dark current, indicating that the PbS QD film has a lower mobility. This is because, in the EPD-formed films, charge carriers not only need to accomplish QD-QD hopping but also need to accomplish Cluster – Cluster hopping.

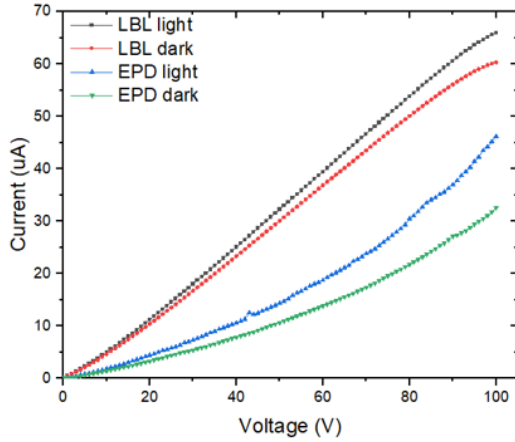


Fig. 6. The dark and the illuminated current of devices prepared by LBL and EPD way. The light source is a NIR laser at 1310 nm with intensity of 16 $\mu\text{W}/\text{cm}^2$.

C. Responsivity Characterization of Photoconductors

Responsivity (R) can more intuitively reflect the sensitivity of the photodetector to light, and it can be calculated using the following equation:

$$R = \frac{I_p - I_d}{P_{in}} \quad (1)$$

Where I_p is the photocurrent generated by the photodetector under illumination, and P_{in} is the incident optical power.

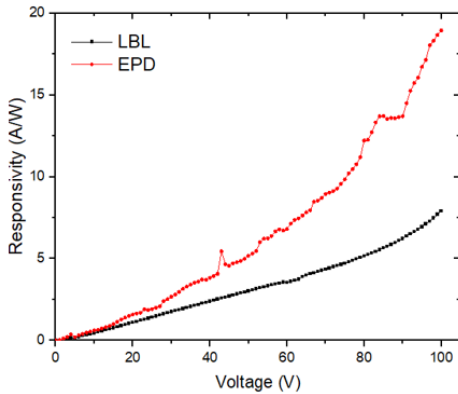


Fig. 7. The Responsivity of devices prepared by LBL and EPD way. Illuminated under 1310 nm, 16 $\mu\text{W}/\text{cm}^2$.

The Responsivity-Voltage (R-V) curve is shown in Figure 7. Within the range indicated in the figure, the photodetector fabricated using the EPD method exhibits higher responsivity. This could be because the PbS QDs in the film formed by the EPD method have shorter spacing and higher long-range effectiveness [9]. This allows more charge carriers to be generated under illumination and stably transported to both electrodes of the photoconductor.

IV. CONCLUSION

In summary, a high responsivity photoconductor is achieved and the fabrication of PbS QD films using EPD method is fulfilled. The SEM image show the PbS QD film having a more regular arrangement than the film using LBL method. The optimized responsivity reaches 18.95 A/W, which is an enhancement compared with the LBL method. Our work shows a simple and productive method of optimizing the responsivity of photoconductor basing PbS QDs. This shows that using the EPD method to prepare PbS QD films helps to enhance various performance aspects of photoconductors. This method will provide indispensable assistance for the advancement and expansion of the near-infrared photodetector market.

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